



Time = 15 Minutes

Instructor Solution - Quiz No 3

[20 points]

Note - Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

Question 1 [08 points] Counting

From 5 seniors, 4 juniors, 1 sophomore and 8 freshmen, in how many ways can the group of five be chosen if there must be at least one member from each class?

Question 2 [12 points] Calculating Probability using Counting

Given that there were 4 heads in 9 tosses, what is the probability that the first three tosses were heads? Assume it is a biased coin with $P(H) = p$.

You are required to solve this using **ONLY** the counting method and using the following two events. You need to back your working with solid reasoning. This should definitely include the reasoning, how can the required probability be calculated using the counting methods.

- event A: the first 3 tosses were heads.
- event B: 4 out of 9 tosses were heads.

Note - $P(A | B) = \frac{P(A \cap B)}{P(B)} =$